

Jordan Mark Barbone, M.A.

DATA SCIENTIST, R DEVELOPER

West Chester, PA

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R programmer with experience in statistical programming, monitoring, and report automation in cognitive and CNS clinical trials

For an updated version see [jmbarbone/barbone-cv](#)

Skills

R - Advanced	package development, data analysis, statistical programming, data visualization, ETL/Data integration, <code>{tidyverse}</code> , <code>Rmarkdown</code> , <code>quarto</code>
Other programming experience	Intermediate: <code>Python</code> (<code>{pyspark}</code>) Basic: <code>SQL</code> , <code>git</code> , <code>bash</code>
Clinical trials	clinical data management programming, centralized monitoring, eCOA, data review, Rater training and qualification
Psychology	behavioral statistics, research methods, cognitive psychology, learning psychology

Experience

Data Scientist

Jun 2021 - Present

COGSTATE

New Haven, CT (Remote)

- Independently developed internal **R packages** to advance programming, visualization, normative data management, central monitoring, data access, and general utilities
- Designed and development **R shiny** applications for displaying central monitoring activity and performing interactive analyses
- Refactored Data Engineering **PySpark** code into a **python** package for data transformations within **Azure Databricks** and **DeltaTable** management
- Refactored legacy **SQL** code to seamlessly integrate **PySpark** functionalities and enforce best coding practices
- Designed and implemented an efficient process for managing and updating and tracking normative data, collaborating closely with Science and Product Management teams
- Authored comprehensive standards and protocols for **Jira** workflows and **git** management, ensuring consistency and efficiency
- Developed and managed a robust daily and weekly reporting system with **R tidyverse** scripts for the monitoring of a high-profile international Alzheimer's Disease trial
- Analyzed and interpreted requirements from Science and Data Management to drive strategic advancements within our Data Lakehouse
- Developed, reported, and presented analyses with **R tidyverse** scripts for custom data queries related to both ongoing and concluded clinical, providing actionable insights for rater training and central monitoring
- Developed a **python** data extraction, reformatting, and storage processes for academic research studies, greatly enhancing accessibility for end-users
- Contributed to the creation of poster materials for industry conferences and presentations, showcasing research, developments, and services

Principal Data Analyst

Dec 2019 - Jun 2021

MADRIGAL PHARMACEUTICALS

Conshohocken, PA

- Developed internal **R packages** with **tidyverse** features for the creation of daily metric and reports
- Aggregated and organized data sources in STDm, ADaM, TFLS, and pre-finalized formats; compiled source data from multiple vendor data bases and reports; reconciled differences between databases using **R tidyverse** suite
- Analyzed efficacy, safety, and exploratory data for: ongoing phase 3 studies in NASH and NAFLD; ad hoc analyses for phase 2; biomarker and patient reported outcomes with **R**
- Reviewed and reported on statistical testing in **R** such as: ANCOVA, LS-Means, ROC, Kruskal-Wallis, Wilcoxon rank sum, t-test, correlations (Pearson product, Spearman rho, Kendall's tau)
- Trained and onboarded data analyst

Clinical Data Scientist

Jan 2019 - Nov 2019

SIGNANT HEALTH (FORMERLY BRACKET AND CRF HEALTH)

Wayne, PA

- Developed analytic **R** scripts using **tidyverse** features for clinical trial operational and data management monitoring.
- Created **R** scripts for processing clinical trial outcome data for risk-based statistical quality monitoring
- Provided analytic support for internal and external research projects, including conference poster preparation

Clinical Data Specialist

Jan 2018 - Dec 2018

SIGNANT HEALTH (BRACKET)

Wayne, PA

- Analyzed central monitoring data for clinical trials using **R** scripts
- Provided analytic support for internal and external research projects, including conference poster preparation
- Reviewed audio data from ADAS-Cog, MMSE, and ADCS-ADL administrations; provided feedback on standard administration and scoring, and submitted score change reviews for clinical trial sites

Clinical Associate

May 2016 - Dec 2017

SIGNANT HEALTH (BRACKET)

Wayne, PA

- Reviewed clinical trial raters' qualifications and recommended standardized training programs.
- Developed a training manual in **Microsoft Word** for the standardized administration of ADAS-Cog and MMSE within clinical trials, along with additional training presentations in **Microsoft PowerPoint**, ensuring alignment with company standards
- Reviewed audio data from ADAS-Cog, MMSE, and ADCS-ADL administrations; provided feedback on standard administration and scoring, and submitted score change reviews for clinical trial sites
- Developed new **Excel** trackers for central review tracking for individual central reviewers; compiled reports and presented information for central reviewer management

Clinical Assistant

Aug 2015 - May 2016

SIGNANT HEALTH (BRACKET)

Wayne, PA

- Reviewed clinical trial raters' qualifications and recommended standardized training programs
- Scheduled re-training appointments for clinical trial raters in collaboration with central quality reviewers
- Created and maintained study trackers and teleconference schedules in **Excel**

Writing Tutor

Aug 2014 - May 2015

LEARNING ASSISTANCE & RESOURCE CENTER

West Chester University of Pennsylvania,

West Chester, PA

- **Tutored** students in one-on-one or small group sessions to enhance writing skills and promote independent, active learning
- Monitored the effectiveness of the Academic Development Program and tracked student progress by maintaining daily activity logs, using ****Bloom's Taxonomy****, and preparing biweekly student reports and semester evaluations

Education

Master of Arts

West Chester, PA

WEST CHESTER UNIVERSITY OF PENNSYLVANIA

Aug 2017 - May 2019

- General Psychology, Advisor: Geeta Shivde, Ph.D.
- Thesis: The effects of participant-selected background music on executive function task performance
- GPA: 4.00

Non-degree Graduate Student

West Chester, PA

WEST CHESTER UNIVERSITY OF PENNSYLVANIA

Aug 2016 - Jun 2017

- Classes in Statistics, Research Methods, and Psychometrics
- GPA: 4.00

Bachelor of Arts

West Chester, PA

WEST CHESTER UNIVERSITY OF PENNSYLVANIA

Aug 2011 - May 2015

- Major in Psychology, Minor in Biology
- Cumulative GPA: 3.470
- Major GPA: 3.834

CRAN packages

Barbone, J. M. (2024). *Fuj: Functions and utilities for jordan*. <https://jmbarbone.github.io/fuj/>

Barbone, J. M. (2024). *Mark: Miscellaneous, analytic r kernels*. <https://github.com/jmbarbone/mark>

Barbone, J. M., & Garbuszus, J. M. (2024). *openxlsx2: Read, write and edit xlsx files*. <https://janmarvin.github.io/openxlsx2/>

Barbone, J. M. (2023). *Echo: Echo code evaluations*. <https://github.com/jmbarbone/echo>

Barbone, J. M. (2023). *Scribe: Command argument parsing*. <https://jmbarbone.github.io/scribe/>

Papers

PEER REVIEWED PUBLICATIONS

Younossi, Z. M., Stepanova, M., Taub, R. A., **Barbone, J. M.**, & Harrison, S. A. (2021). Hepatic fat reduction due to resmetirom in patients with nonalcoholic steatohepatitis is associated with improvement of quality of life. *Clinical Gastroenterology and Hepatology*. <https://doi.org/10.1016/j.cgh.2021.07.039>

Solomon, T. M., **Barbone, J. M.**, Feaster, H. T., Miller, D. S., deBros, G. B., Murphy, C. A., & Michalczuk, D. (2019). Comparing the standard and electronic versions of the Alzheimer's disease assessment scale—cognitive subscale: A validation study. *The Journal of Prevention of Alzheimer's Disease*, 6(4), 237–241. <https://doi.org/10.14283/jpad.2019.27>

Irani, F., **Barbone, J. M.**, Beausoleil, J., & Gerald, L. (2017). Is asthma associated with cognitive impairments? A meta-analytic review. *Journal of Clinical and Experimental Neuropsychology*, 39(10), 965–978. <https://doi.org/10.1080/13803395.2017.1288802>

MASTER'S THESIS

Barbone, J. M. (2019). *The effects of participant-selected background music on executive function task performance* [Master's thesis, West Chester University of Pennsylvania]. <https://doi.org/10.13140/RG.2.2.34109.41446>

Posters and presentations

COGSTATE - DIGITAL COGNITION

Tronchin, G., Kingery, L., Leventhal, R., Bartolic, E., Wacker, S., **Barbone, J. M.**, & Wessels, A. M. (2024, March). Central monitoring of rater performance and characteristics of efficacy and assessments in the TRAILBLAZER-ALZ 2 study. *Poster Presented at the International Conference on Alzheimer's and Parkinson's Disease (ADPD), Lisbon, Portugal, and Online*.

Barbone, J. M., Khurshid, K., Edgar, C. J., & Maruff, P. (2023, July). Equating the ADAS-Cog and MMSE to the Cogstate Brief Battery scores. *Poster Presented at the Alzheimer's Association International Conference (AAIC), Amsterdam, Netherlands*. <https://doi.org/doi/10.1002/alz.077060>

Edgar, C. J., Maruff, P., Harrison, J. E., **Barbone, J. M.**, Leventhal, R., & Alam, J. j. (2022). Validity and reliability of a composite cognitive outcome measure for clinical trials in dementia with lewy bodies. *Poster Presented at the ADPD Conference, Barcelona, Spain*. Video: <https://youtu.be/N8TbGlmGV5U>. <https://www.cogstate.com/wp-content/uploads/2022/04/00171-Validity-and-reliability-of-a-composite-cognitive-outcome-in-d273-Read-Only.pdf>

Wacker, S., & **Barbone, J. M.** (2022). Rater perspectives on Applied Training of cognitive clinical outcome assessments, delivered by Neuropsychology experts. *Poster Presented at AAIC, San Diego, California, USA*. <https://www.cogstate.com/wp-content/uploads/2022/08/Rater-Perspectives-on-Applied-Training-of-Cognitive.pdf>

Edgar, C. J., Amariglio, R., **Barbone, J. M.**, Chandler, J., Coons, S., Donohue, M., Lenderking, W., & Sperling, R. (2022, December). Preliminary evidence for reliability and validity of the Interpersonal Functioning and Daily Activities Questionnaire (IFDAQ) in the A4/LEARN pre-randomization sample. *Slides Presented at Clinical Trials on Alzheimer's Disease (CTAD) Conference, San Francisco, California, USA*.

MADRIGAL - LIVER DISEASE

Harrison, S. A., Taub, R., **Barbone, J. M.**, Franc, J., & Karsdal, M. A. (2020, March). Resmetirom, a beta selective thyroid hormone receptor agonist, reduces net collagen III deposition in nonalcoholic Steatohepatitis. *Poster to Have Been Presented at American Association for the Study of Liver Diseases (AASLD) Emerging Topic Conference 2020, Nuclear Receptors in Nonalcoholic Fatty Liver Diseases (Conference Cancelled)*.

Harrison, S. A., Taub, R. A., Karsdal, M. A., Franc, J., Bashir, M., **Barbone, J. M.**, Neff, G., Gunn, N. T., & Moussa, S. (2020, November). Algorithm for predicting advanced NASH fibrosis on screening biopsy in resmetirom phase 3 MAESTRO-NASH clinical trial. *Poster Presented at the AASLD Liver Meeting Digit Experience Conference*.

Younossi, Z. M., Stepanova, M., Taub, R. A., **Barbone, J. M.**, Moussa, S., & Harrison, S. A. (2020, November). Improvement of health-related quality of life is associated with improvement of fat fraction by MRI-PDFF in patients with nonalcoholic steatohepatitis treated with resmetirom. *Poster Presented at the AASLD Liver Meeting Digit Experience Conference*.

SIGNANT HEALTH (BRACKET) - ECOA DATA QUALITY

Crittenden, K., Machizawa, S., Feaster, H. T., **Barbone, J. M.**, Hong, B., Verma, P., & Zhou, W. (2020, April). Cross-cultural differences in PANSS item ratings: Comparisons of six geo-cultural regions. *Poster Presented at the 2020 Schizophrenia International Research Society (SIRS) Conference, Florence, Italy*. <https://doi.org/10.1093/schbul/sbaa031.103>

Roy, M., Brown, J., Hong, B., Benecke, R., Chen-Tackett, Z., Zhou, W., **Barbone, J. M.**, Feaster, H. T., & Sachs, G. (2020, September). eCOA prompted MADRS interview: Balancing through assessment and efficiency. *Poster Presented at ISCTM*.

Feaster, H. T., **Barbone, J. M.**, Miller, D. S., & Solomon, T. M. (2019, June). Rater remediation on the ADAS-cog leads to longitudinal improvement in clinical trial data quality. *Poster Presented at the Alzheimer's Association International Conference (AAIC), Los Angeles, CA, USA*. <https://doi.org/10.1016/j.jalz.2019.06.3923>

Karas, S. M., **Barbone, J. M.**, Solomon, T. M., Hong, B., & Feaster, H. T. (2019, June). Impact of data quality programs on MMSE inclusion criteria in Alzheimer's disease clinical research trials. *Poster Presented at the Alzheimer's Association International Conference (AAIC), Los Angeles, CA, USA*. <https://doi.org/10.1016/j.jalz.2019.06.862>

Solomon, T. M., Feaster, H. T., Karas, S. M., Garcia-Valdecasas, M., **Barbone, J. M.**, DiGregorio, D. T., DeBonis, D., & Miller, D. S. (2019, July). The evolution of technology in data quality programs for Alzheimer's disease clinical trials: What have we learned and where are we going. *Poster Presented at the Alzheimer's Association International Conference (AAIC), Los Angeles, CA, USA*. <https://doi.org/10.1016/j.jalz.2019.06.4332>

Seichepine, D. R., Solomon, T. M., Jacobs, E. B., **Barbone, J. M.**, DiGregorio, D. T., & Miller, D. S. (2018, February). Comparison between "flat" and "enhanced" eCOA of MMSE Attention and Calculation in clinical trials of Alzheimer's disease. *Presented at the ISCTM 14th Annual Scientific Meeting, Washington, D.C., USA*. <https://doi.org/10.13140/RG.2.2.16193.61289>

Solomon, T. M., **Barbone, J. M.**, Miller, D. S., & Feaster, H. T. (2018, June). Pilot validation of an Electronic Alzheimer's Disease Assessment Scale - Cognitive Subscale (eADAS-Cog). *Poster Presented at the 2018 Alzheimer's Association International Conference, Chicago, IL, USA*. <https://doi.org/10.1016/j.jalz.2018.06.719>

Feaster, H. T., **Barbone, J. M.**, Garcia-Valdecasas-Colell, M., & Solomon, T. M. (2018, July). Electronic ADAS-Cog (eADAS-Cog) data quality: How do countries compare? *Poster Presented at the 2018 Alzheimer's Association International Conference, Chicago, IL, USA*. <https://doi.org/10.1016/j.jalz.2018.06.1373>

Karas, S. M., **Barbone, J. M.**, DeBonis, D., & Solomon, T. M. (2018, July). The use of statistical modeling to complement data quality programs. *Poster Presented at the 2018 Alzheimer's Association International Conference, Chicago, IL, USA*. <https://doi.org/10.1016/j.jalz.2018.06.066>

Solomon, T. M., Karas, S. M., **Barbone, J. M.**, DiGregorio, D. T., Miller, D. S., & Feaster, H. T. (2018, July). Analysis of the rates and types of errors on the enhanced eCOA version of the Alzheimer's Disease Assessment Scale - Cognitive subscale and Mini-mental State Examination used in dementia clinical trials. *Poster Presented at the 2018 Alzheimer's Association International Conference, Chicago, IL, USA*. <https://doi.org/10.1016/j.jalz.2018.06.070>

Barbone, J. M., Solomon, T. M., Feaster, H. T., Garcia-Valdecasas-Colell, M., & Miller, D. S. (2018, October). MMSE screening data quality for Alzheimer's disease studies across countries. *Presented at the 11th Clinical Trials on Alzheimer's Disease (CTAD), Barcelona, Spain*. <https://doi.org/10.13140/RG.2.2.27921.68966>

- Garcia-Valdecasas-Colell, M., **Barbone, J. M.**, Solomon, T. M., Feaster, H. T., & Tott, N. (2018, October). Identifying impact of rater change on MMSE and CDR data in multi-national Alzheimer's disease clinical trials. *Presented at the 11th Clinical Trials on Alzheimer's Disease (CTAD), Barcelona, Spain.* <https://doi.org/10.13140/RG.2.2.21210.80327>
- Wessels, A. M., **Barbone, J. M.**, DiGregorio, D. T., Miller, D. S., Mullen, J. A., Sims, J. R., & Solomon, T. M. (2018, October). Lanabecestat- Rater performance and error characteristics of efficacy assessments in the DAYBREAK-ALZ study. *Presented at the 11th Clinical Trials on Alzheimer's Disease (CTAD), Barcelona, Spain.* <https://bit.ly/32p18au>
- Butler, A., Feaster, H. T., Miller, D. S., & **Barbone, J. M.** (2017, May). Using iterative user experience design to improve electronic clinical outcomes assessment data quality. *Presented at the ISPOR 22nd Annual International Meeting, Boston, MA, USA.* <https://bit.ly/3isANeP>
- Feaster, H. T., Solomon, T. M., Abi-Saab, D., Vogt, A., **Barbone, J. M.**, Harrison, J., & Miller, D. S. (2017, July). The impact of electronic clinical outcome assessments (eCOA) on Alzheimer's disease clinical trial data quality. *Poster Presented at the Alzheimer's Association International Conference, London, England, UK.* <https://doi.org/10.1016/j.jalz.2017.06.1828>
- Enger, B., & **Barbone, J. M.** (2017, October). Distribution of performance and incorrect ratings in qualification video scoring. *Poster Presented at CNS Summit, Boca Raton, FL, USA.* <https://doi.org/10.13140/RG.2.2.14096.46088>
- Solomon, T. M., **Barbone, J. M.**, Karas, S. M., & Feaster, H. T. (2017, October). Longitudinal impact of audio review on data quality. *Poster Presented at Clinical Trials on Alzheimer's Disease Conference (CTAD), Boston, MA, USA.* <https://bit.ly/2DYaFtd>
- Solomon, T. M., Feaster, H. T., **Barbone, J. M.**, & Miller, D. S. (2017, November). Utilizing audio review to improve ADCS-ADL data quality. *Poster Presented at Clinical Trials on Alzheimer's Disease Conference (CTAD), Boston, MA, USA.* <https://bit.ly/3m1Ku0c>

WCUPA - ASTHMA AND MUSIC COGNITIVE

- Barbone, J. M.**, & Shivde, G. (2019, May). The effects of participant selected background music on executive task performance. *Poster Presented at the Association for Psychological Science, Washington, D.C., USA.* <https://doi.org/10.13140/RG.2.2.26125.79842>
- Barbone, J. M.** (2018, November). Trends in self-reported use of music while studying: Implications for research. *Poster Presented at the Research and Creative Activity Day at West Chester University, West Chester, PA, USA.* <https://doi.org/10.13140/RG.2.2.14075.57124>
- Barbone, J. M.**, Saxena, S., & Irani, F. (2015, February). Neuropsychological performance in individuals with asthma: Two meta-analyses. *Poster Presented at the Eastern Psychological Association Conference, Philadelphia, PA, USA.* <https://bit.ly/3hu6J0i>
- Mulligan, R. D., **Barbone, J. M.**, Saxena, S., Carey-Jr, D., Clappsy, J., & Irani, F. (2015, March). Asthma cognition data collection. *Poster Presented at the West Chester University Psychology Research Day, West Chester, PA, USA.* <https://bit.ly/2ZAD5kn>
- Barbone, J. M.**, Saxena, S., & Irani, F. (2015, July). Does asthma effect neuropsychological performance? A meta-analysis. *Poster Presented at the American Psychological Association Convention, Toronto, Canada.*
- Irani, F., **Barbone, J. M.**, & Beausoleil, J. (2015, August). The impact of asthma on cognitive functioning. *Paper Presented at the American Psychological Association Convention, Toronto, Canada.*